Differences in Differences

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Roadmap

1. Theory: Potential outcomes, identification, key quantities
2. Randomization
   - difference in means, variance estimation
   - covariate adjustment
   - blocking
   - cluster randomization
3. Selection on Observables
   - sub-classification
   - matching (on X)
   - weighting (on X)
   - matching and weighting on $Pr(D = 1)$ (propensity scores)
   - regression
   - sensitivity
4. Difference in Differences
5. Instrumental Variables
6. Regression Discontinuity
7. More panel methods (time permitting)

Weakening or Avoiding Selection on Observables

SOO/CI has become the standard approach across many disciplines

- The common formula even now in many fields: “control for stuff”, say you know there may be confounders, but then treat the result as tentatively correct anyway.

Much of the focus on causal inference starts by noting the insufficiency of SOO/CI and looking for alternatives.

- A (relatively small) number of identification strategies have gained popularity as alternatives: DID, IV, RDD, and some time-series approaches
- These identification strategies are often associated with the “credibility revolution”, but they are not “more causal” than SOO/CI – they just live on different assumptions.
- As with matching, you’ll find there is some degree of inappropriate faith in these techniques because of this association
- Your job is to see that by understanding the assumptions on which these approaches rest.

DID in the Time of Cholera

- Prevailing 19th century wisdom was that cholera was spread by “bad air”
- John Snow, a London physician thought otherwise
- In 1849, the Southwark and Vauxhall water company and the Lambeth company got water from the same polluted part of the Thames in central London. Snow assessed cholera mortality in both areas.
- In 1852, Lambeth moved its waterworks up stream.
- Shortly after, cholera death rates fell sharply in areas served by Lambeth, but not area served by S & V.

Intuitions:

- Why not just compare the two in 1852?
- Why not just compare Lambeth’s service area post-minus-pre?

Two Groups, Two Periods

Definition

Two groups:

- $D = 1$ for (eventually) treated units
- $D = 0$ for control units

Two periods:

- $T = 0$ Pre-Treatment period
- $T = 1$ Post-Treatment period

Potential outcomes $Y_{di}(t)$

- $Y_{1i}(t)$ potential outcome unit i attains in period t when treated between t and $t - 1$
- $Y_{0i}(t)$ potential outcome unit i attains in period t when non-treated

Two Groups and Two Periods

Definition

Causal effect for unit i at time t is

$$\tau_{it} = Y_{1i}(t) - Y_{0i}(t)$$

Observed outcomes $Y_i(t)$ are realized as

$$Y_i(t) = Y_{1i}(t) \cdot (D_i * (t = 1)) + Y_{0i}(t) \cdot (1 - (D_i * (t = 1))$$

Estimand (ATT post-treatment)

$$\tau_{ATT} = \mathbb{E}[Y_{1i}(1) - Y_{0i}(1) | D_i = 1]$$

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{1i}(1)|D_i = 1] - \mathbb{E}[Y_{0i}(1)|D_i = 1]$$

Observed Moments:

	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1)\|D = 1]$	$\mathbb{E}[Y_0(0)\|D = 1]$
Control D=0	$\mathbb{E}[Y_0(1)\|D = 0]$	$\mathbb{E}[Y_0(0)\|D = 0]$

Two Groups and Two Periods

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Two Groups and Two Periods

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	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1)\|D = 1]$	$\mathbb{E}[Y_0(0)\|D = 1]$
Control D=0	$\mathbb{E}[Y_0(1)\|D = 0]$	$\mathbb{E}[Y_0(0)\|D = 0]$

Problem

Missing potential outcome: $\mathbb{E}[Y_0(1)|D = 1]$, ie. what is the average non-treatment outcome in the post-treatment period, for the treated?

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{1i}(1)|D_i = 1] - \mathbb{E}[Y_{0i}(1)|D_i = 1]$$

Observed Moments:

	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1)\|D = 1]$	$\mathbb{E}[Y_0(0)\|D = 1]$
Control D=0	$\mathbb{E}[Y_0(1)\|D = 0]$	$\mathbb{E}[Y_0(0)\|D = 0]$

ID Strategy 1 (not DID): Over-time ("Post-minus-Pre") Comparison

- Estimand: $\mathbb{E}[Y_1(1)|D = 1] - \mathbb{E}[Y_0(0)|D = 1]$
- Works if: $\mathbb{E}[Y_0(0)|D = 1] = \mathbb{E}[Y_0(1)|D = 1]$
- What does this assumption mean?

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{1i}(1)|D_i = 1] - \mathbb{E}[Y_{0i}(1)|D_i = 1]$$

Observed Moments:

	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1)\|D = 1]$	$\mathbb{E}[Y_0(0)\|D = 1]$
Control D=0	$\mathbb{E}[Y_0(1)\|D = 0]$	$\mathbb{E}[Y_0(0)\|D = 0]$

ID Strategy 2 (not DID): Cross-Sectional Comparison (in Post)

- Use: $\mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(1)|D = 0]$
- Assumes: $\mathbb{E}[Y_0(1)|D = 0] = \mathbb{E}[Y_0(1)|D = 1]$
- What does this mean?

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = \mathbb{E}[Y_{1i}(1)|D_i = 1] - \mathbb{E}[Y_{0i}(1)|D_i = 1]$$

Observed Moments:

	Post-Period (T=1)	Pre-Period (T=0)
Treated D=1	$\mathbb{E}[Y_1(1)\|D = 1]$	$\mathbb{E}[Y_0(0)\|D = 1]$
Control D=0	$\mathbb{E}[Y_0(1)\|D = 0]$	$\mathbb{E}[Y_0(0)\|D = 0]$

ID Strategy 3: Difference-in-Differences (DID)

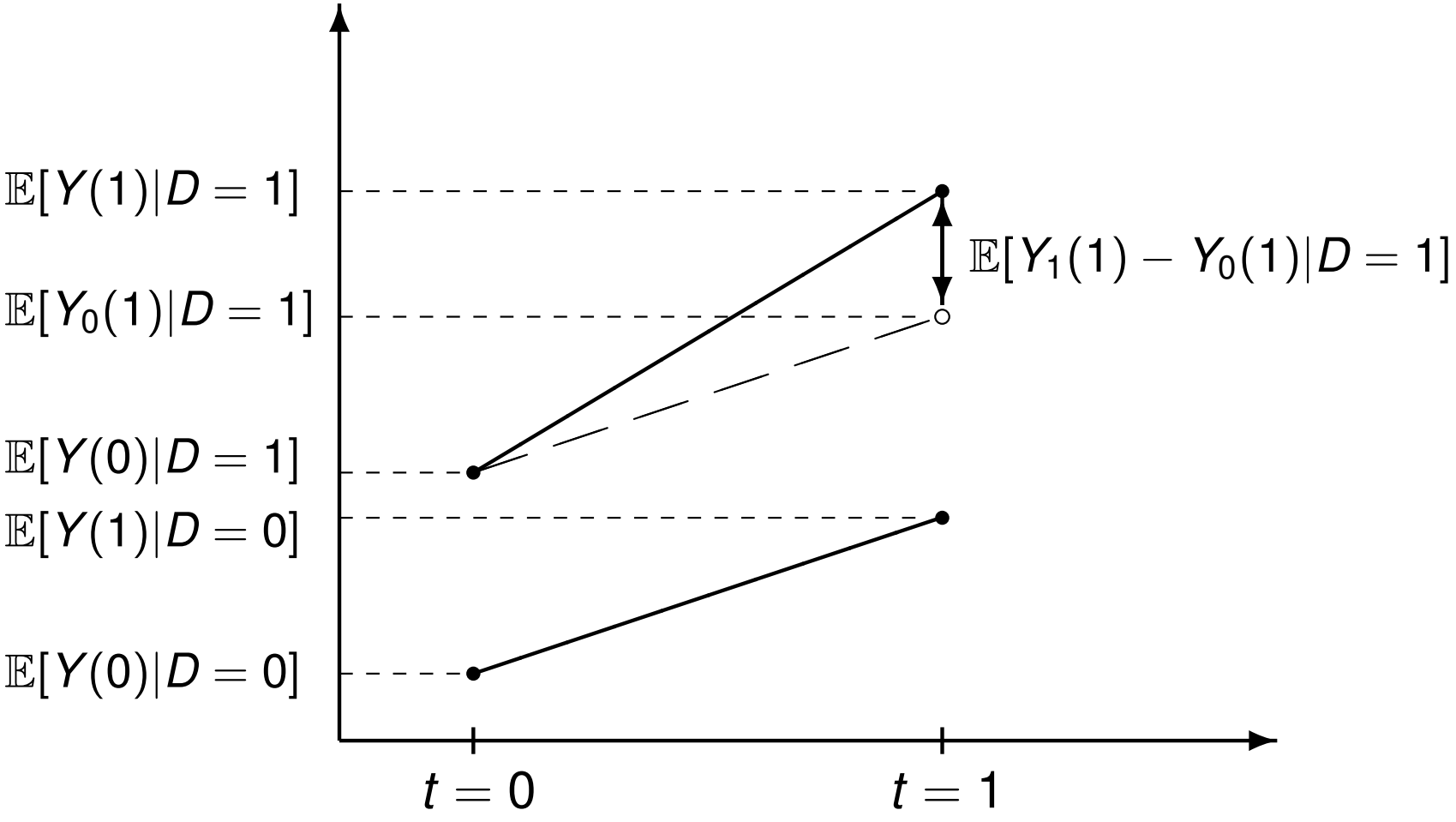
- Use:

$$\mathbb{E}[Y(1) - Y(0)|D = 1] - \mathbb{E}[Y(1) - Y(0)|D = 0]$$

- Assumes: $\mathbb{E}[Y_0(1) - Y_0(0)|D = 1] = \mathbb{E}[Y_0(1) - Y_0(0)|D = 0]$

"Parallel trends" assumption

Graphical Representation: Difference-in-Differences



Identification with Difference-in-Differences

Identification Assumption (Parallel Trends)

Conventionally we write,

$$\mathbb{E}[Y_0(1) - Y_0(0)|D = 1] = \mathbb{E}[Y_0(1) - Y_0(0)|D = 0]$$

But note that this is equivalent to:

$$\Delta Y_0 \perp D$$

where $\Delta Y_0 \equiv \boldsymbol{Y}_0(1) - \boldsymbol{Y}_0(0)$

Identfication Result

Under parallel trends, the ATT is identified by the DID estimator:

$$\begin{aligned}\tau_{DID} &= \{\mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 1]\} - \{\mathbb{E}[Y(1)|D = 0] - \mathbb{E}[Y(0)|D = 0]\} \\ &= \mathbb{E}[Y_1(1) - Y_0(1)|D = 1] = ATT\end{aligned}$$

Proof

$$
\begin{aligned}
\tau_{DID} &= \{\mathbb{E}[Y(1)|D=1] - \mathbb{E}[Y(0)|D=1]\} - \{\mathbb{E}[Y(1)|D=0] - \mathbb{E}[Y(0)|D=0]\} \\
&= \{\mathbb{E}[Y_1(1)|D=1] - \mathbb{E}[Y_0(0)|D=1]\} - \{\mathbb{E}[Y_0(1)|D=0] - \mathbb{E}[Y_0(0)|D=0]\} \\
&= \{\mathbb{E}[Y_1(1)|D=1] - \mathbb{E}[Y_0(0)|D=1]\} - \{\mathbb{E}[Y_0(1)|D=1] - \mathbb{E}[Y_0(0)|D=1]\} \\
&= \mathbb{E}[Y_1(1)|D=1] - \mathbb{E}[Y_0(1)|D=1] \\
&= ATT
\end{aligned}
$$

Another view: additive effects

- Change in control unit's outcome, ΔY_0, is "effect of time", $\Delta Y_0 = \delta_0$
- Change in treated unit's outcome is effect of time and treatment, $\Delta Y_1 = \delta_1 + \tau_{ATT}$
- We estimate $DID = \Delta Y_1 - \Delta Y_0 = (\delta_1 + \tau_{ATT}) - \delta_0$
- ...equals τ_{ATT} if $\delta_0 = \delta_1$

Difference-in-Differences: Estimators

Estimand (DID)

$$\tau_{DID} = \{\mathbb{E}[Y(1)|D=1] - \mathbb{E}[Y(1)|D=0]\} - \{\mathbb{E}[Y(0)|D=1] - \mathbb{E}[Y(0)|D=0]\}$$

Estimator

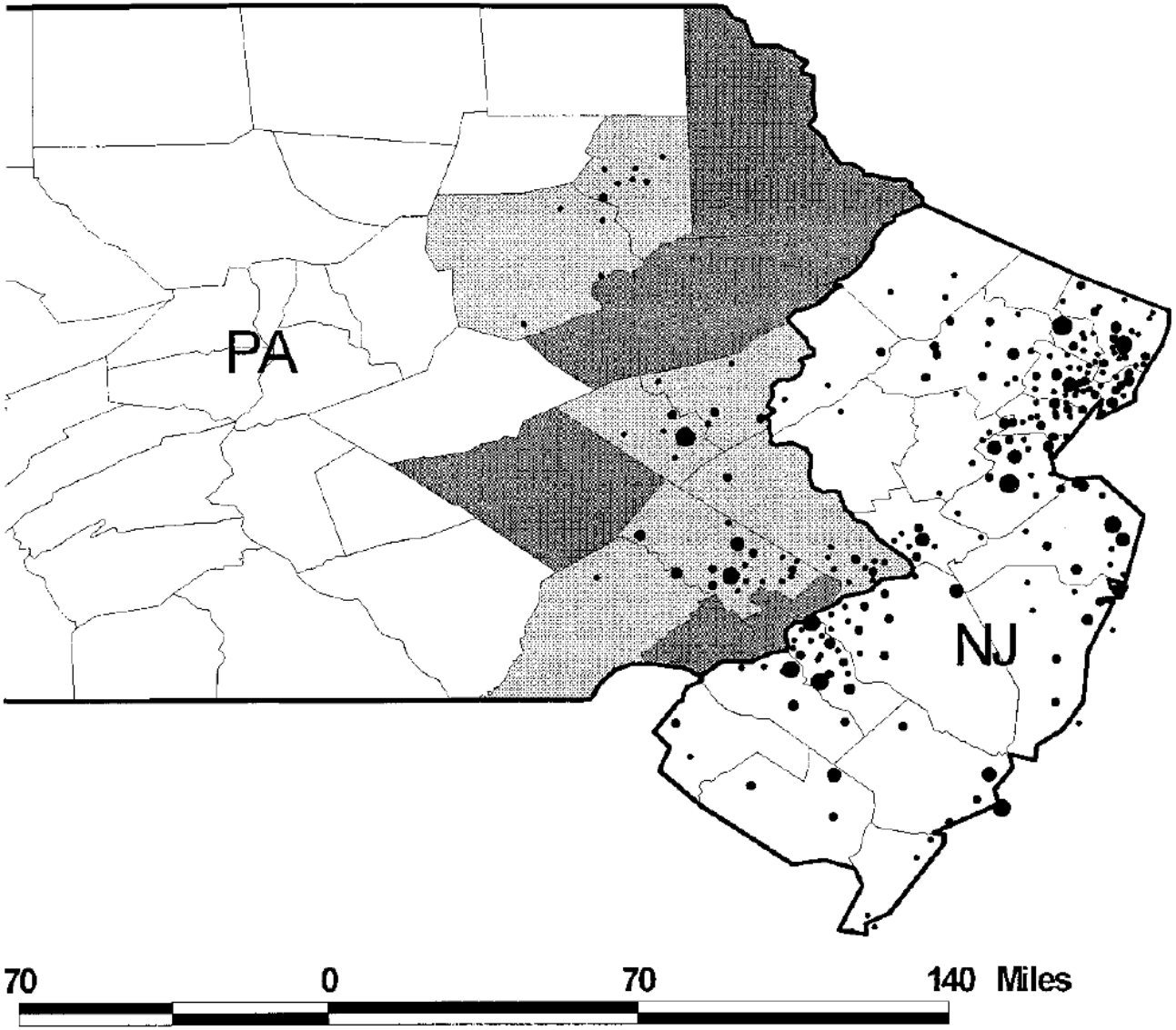
$$\frac{1}{N_1}\sum_{D_i=1}\{Y_i(1) - Y_i(0)\} - \frac{1}{N_0}\sum_{D_i=0}\{Y_i(1) - Y_i(0)\},$$

where N_1 and N_0 are the number of treated and control units respectively.

Example: Minimum wage laws and employment

- Do higher minimum wages decrease low-wage employment?
- Card and Krueger (1994) consider impact of New Jersey's 1992 minimum wage increase from $4.25 to $5.05 per hour
- Compare employment in 410 fast-food restaurants in New Jersey and eastern Pennsylvania before and after the rise
- Survey data on wages and employment from two waves:
  - Wave 1: March 1992, one month before min. wage increase
  - Wave 2: December 1992, eight months after increase

Locations of Restaurants (Card and Krueger 2000)



Sample Means: Minimum wage laws and employment

Variable	Stores by state		
	PA (i)	NJ (ii)	Difference, NJ − PA (iii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)
3. Change in mean FTE employment	−2.16 (1.25)	0.59 (0.54)	2.76 (1.36)

Difference-in-Differences: Estimators

Estimator (Regression: Repeated Cross-Sections)

***In the two period case**, the same estimator can be computed by regression:*

$$Y = \mu + \gamma \cdot D + \delta \cdot T + \tau \cdot (D \cdot T) + \varepsilon,$$

where $\mathbb{E}[\varepsilon | D, T] = 0$.

	After (T=1)	**Before (T=0)**	**After - Before**
Treated D=1	$\mu + \gamma + \delta + \tau$	$\mu + \gamma$	$\delta + \tau$
Control D=0	$\mu + \delta$	μ	δ
Treated - Control	$\gamma + \tau$	γ	τ

Again, parallel trends: "time-effect" δ same for treated and controls

Regression: Minimum wage laws and employment

```
> d <- read.dta("CK1994_longformat.dta",convert.factors = FALSE)
> head(d[, c('ID', 'nj', 'postperiod', 'emptot')])
  ID nj postperiod emptot
1  1  0          0  40.50
2  1  0          1  24.00
3  2  0          0  13.75
4  2  0          1  11.50
5  3  0          0   8.50
6  3  0          1  10.50
```

Regression: Minimum wage laws and employment

```
with(d,
 (
  mean(emptot[nj == 1 & postperiod == 1], na.rm = TRUE) -
  mean(emptot[nj == 1 & postperiod == 0], na.rm = TRUE)
  )  -
  (mean(emptot[nj == 0 & postperiod == 1], na.rm = TRUE) -
  mean(emptot[nj == 0 & postperiod == 0], na.rm = TRUE)
  )
 )
[1] 2.753606
```

Regression: Minimum wage laws and employment

```
> ols <- lm(emptot ~ postperiod * nj, data = d)
> coeftest(ols)

t test of coefficients:

              Estimate Std. Error t value Pr(>|t|)
(Intercept)    23.3312     1.0719 21.7668  < 2e-16 ***
postperiod     -2.1656     1.5159 -1.4286  0.15351
nj             -2.8918     1.1935 -2.4229  0.01562 *
postperiod:nj   2.7536     1.6884  1.6309  0.10331
```

Difference-in-Differences: Threats to Validity

1. **Non-parallel trends**: Often treatments/programs are targeted/selected-into based on pre-existing *trend* in outcomes.
   - "Ashenfelter dip" (more generally, regression to the mean): participants in training programs often treatment *because* of a recent increase in their need for it, and at the same time might be likely to soon show improvement in their outcome even absent treatment just due to mean reversion. (I.e. ΔY_0 differs between groups).
     - e.g. NGOs may target villages that experiences recent setback
     - e.g. Recently unemployed folks more likely to join a job training program.
   - Most directly: If we are here because you don't believe

     $$Y_0 \perp\!\!\!\perp D$$

     ...then what on makes you believe

     $$\Delta Y_0 \perp\!\!\!\perp D$$

   - IOW, think of ΔY_0 as a characteristic of each unit, and ask if you think it could be associated with D.

Checks for Difference-in-Differences Design

Is the parallel-trends assumption testable?

Checks for Difference-in-Differences Design

Two kinds of falsification tests for non-parallel trends

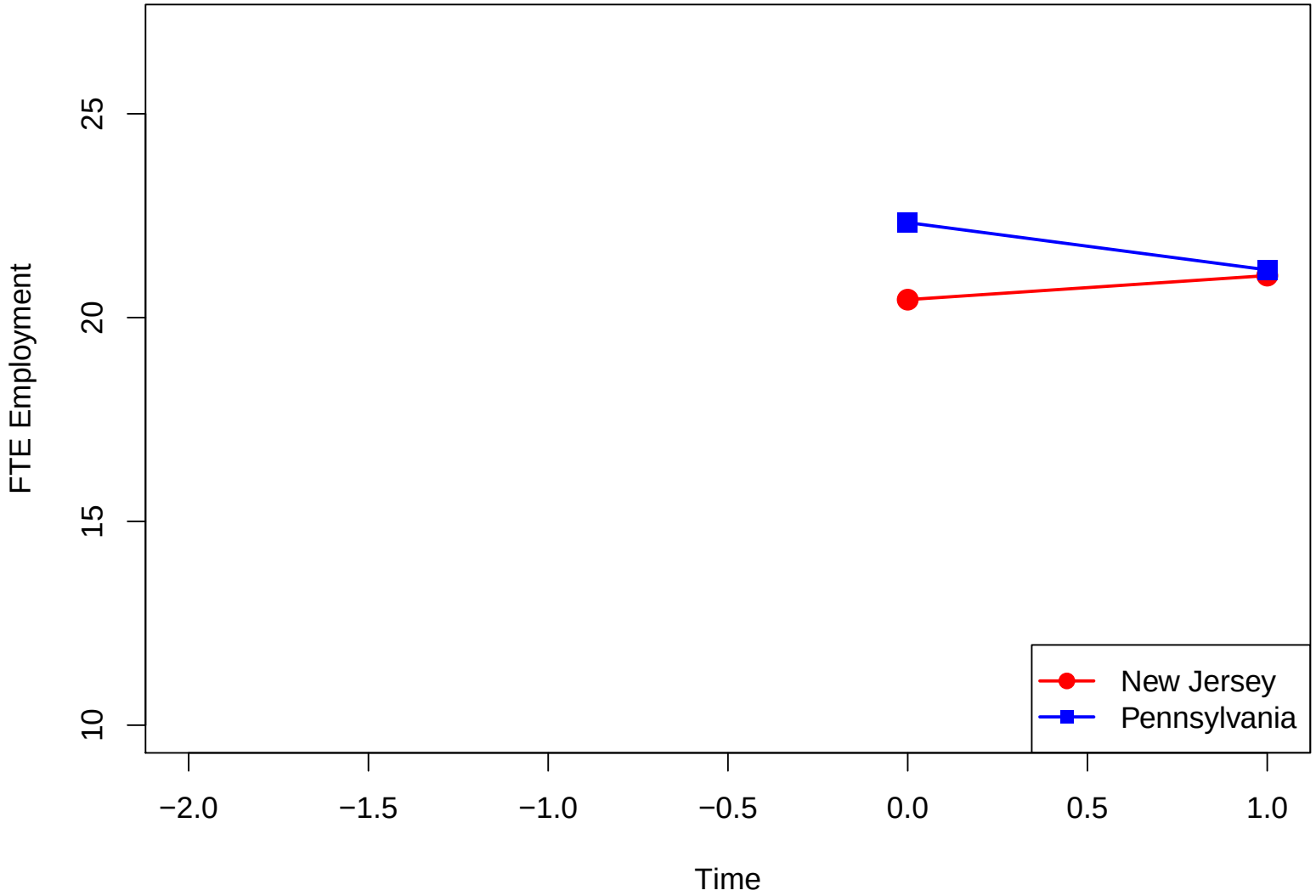
1. Falsification test using data for prior periods
   - Given parallel trends during periods $t = -1, 0, 1$, we have:

$$\mathbb{E}[Y(0) - Y(-1)|D = 1] - \mathbb{E}[Y(0) - Y(-1)|D = 0] = 0$$

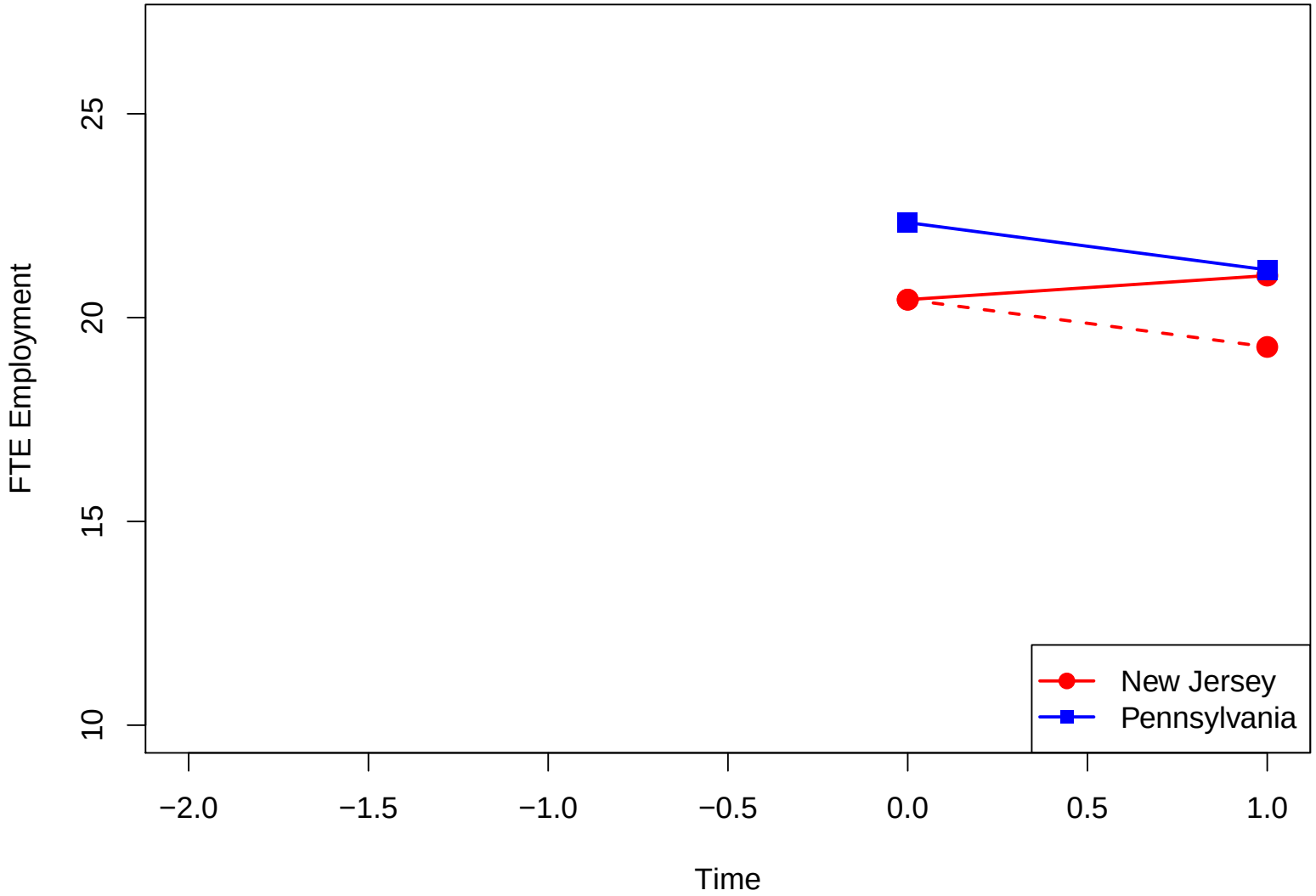
   - run placebo DID on data from $t = -1, 0$.
   - what "treatment effect" should we get?

2. Falsification test using alternative outcome or treated sub-group that is not supposed to be affected by the treatment
   - troubling if DID from the placebo outcome or un-movable group is non-zero.

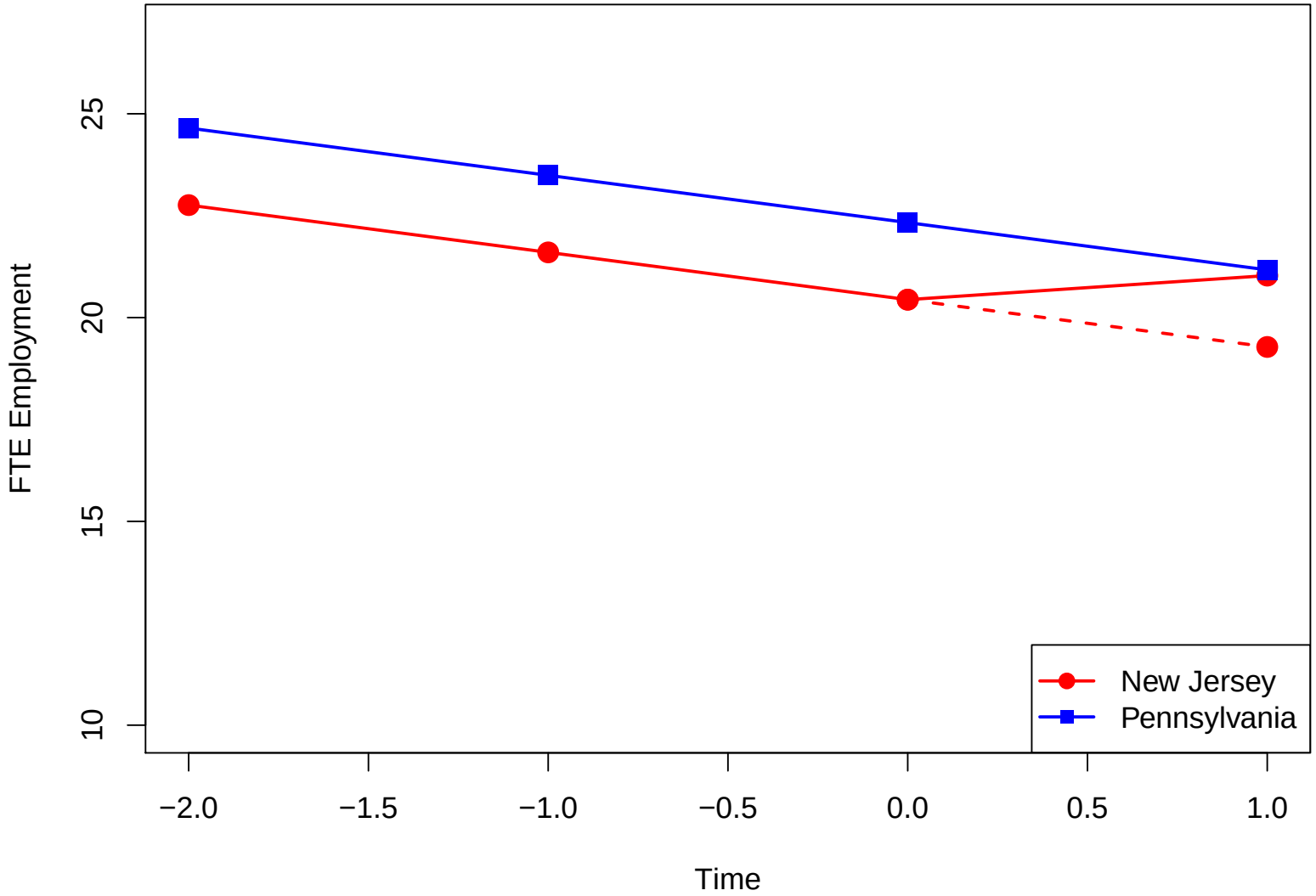
Falsification test: Data for prior periods



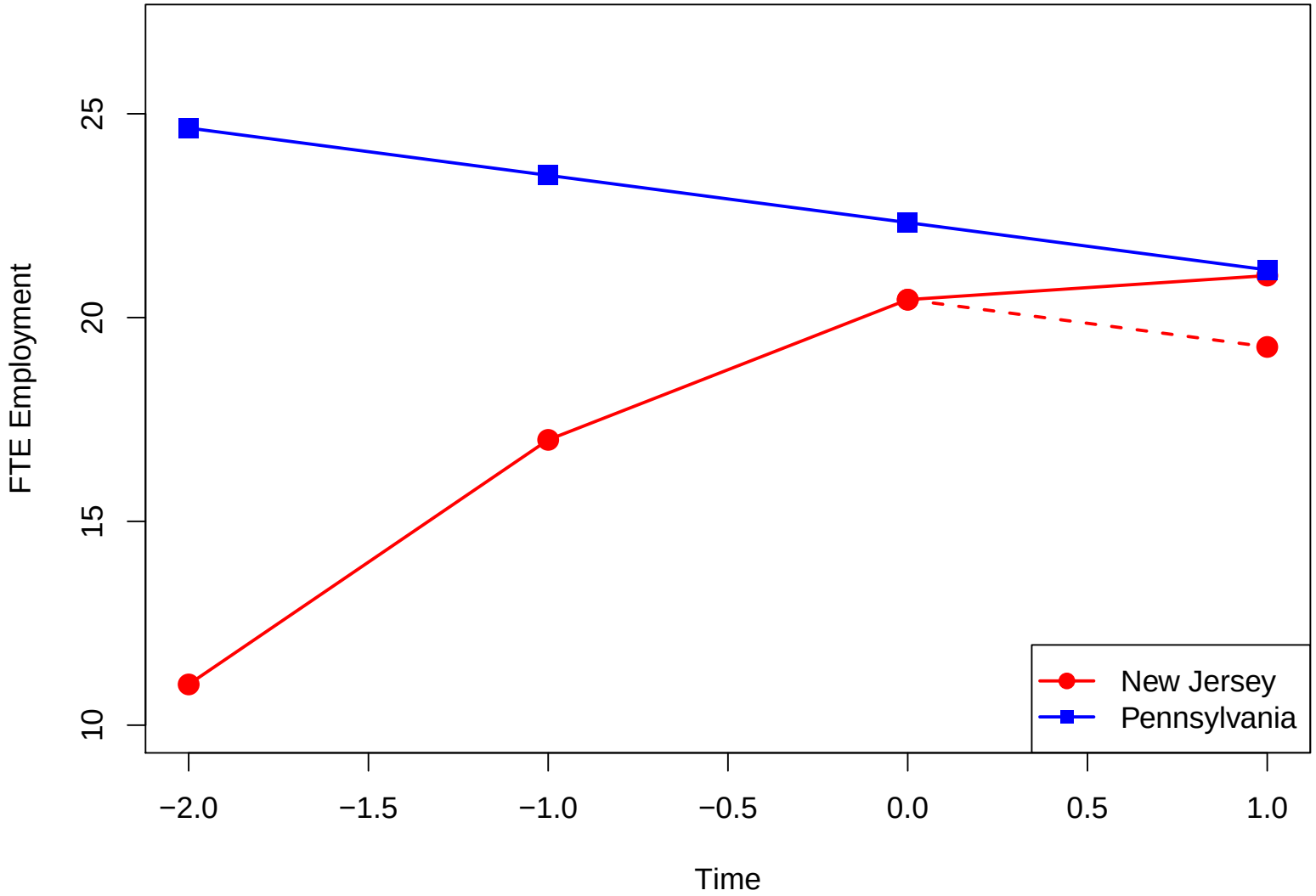
Falsification test: Data for prior periods



Falsification test: Data for prior periods



Falsification test: Data for prior periods



Other approaches to worrying about parallel trends

1. **Sensitivity/ partial ID**

   - Recently-best-known is perhaps Rambachan & Roth (2023)
   - Suggests checking if the effect estimate can survive non-parallelness k times larger than the non-parallelness in the prior trends.
   - A lot of papers don't even survive at $k = 1$, which is clearly bad (Chiu, Lan, Liu, Xu, *forthcoming APSR*)
   - Of course there is no (convincing) rule for "how big of a k is big enough"

There are other (partial) identification strategies that amount to logically arguing about quantities that correspond to the parallel trends assumption (beyond our scope here, but e.g. stability controlled quasi-experiment; using pre-treatment outcome as an imperfect placebo).

Difference-in-Differences: Further Threats to Validity

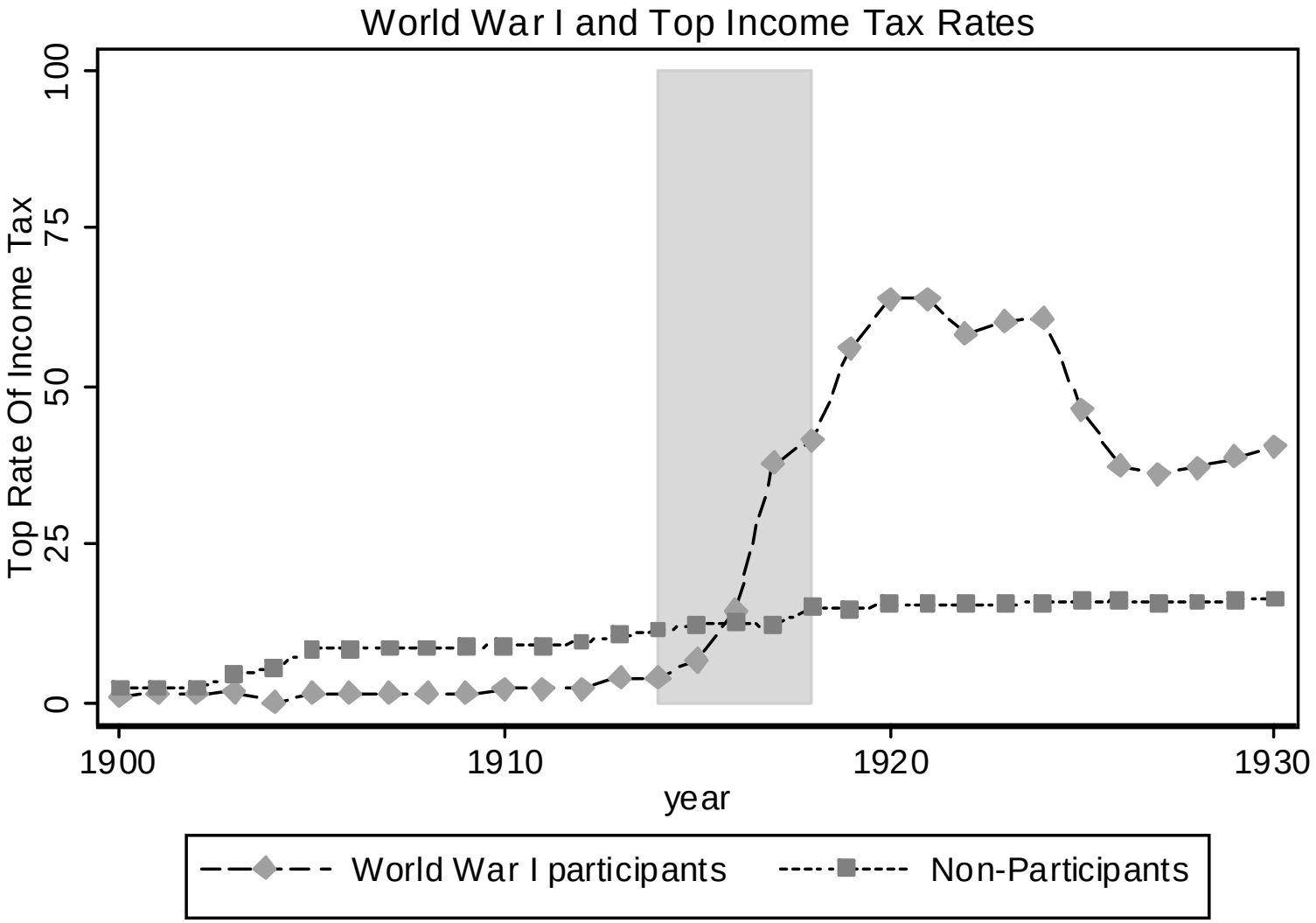
2. **Compositional differences**

- In repeated cross-sections, we do not want the composition of the sample to change between periods.
- Example:
  - Hong (2011) uses repeated cross-sectional data from Consumer Expenditure Survey (CEX) containing music expenditures and internet use for random samples of U.S. households
  - Exploits the emergence of Napster in June 1999 as a natural experiment, comparing internet users and internet non-users, before and after emergence of Napster

Difference-in-Differences: Further Threats to Validity

3. **Long-term effects versus reliability**:
   - Often you may want to know what happens over longer period of times (e.g. effect of polity shift, a conflict, adoption of a policy)
   - But parallel trends assumption is more viable over a short time-window. In the long-run, many other things may happen that could confound the effect of the treatment
   - And short-term effects generally can't be "extrapolated" to estimate long-term effects

Effect of War on Tax Rates (Scheve & Stasavage 2010)



Difference-in-Differences: Further Threats to Validity

4. **Functional form dependence**: The differencing is not-invariant to changes of scale. Magnitude or sign of the DID effect may be sensitive to this.
   - Higher risk when average outcomes for controls and treated are very different at baseline.
   - Example: Cash transfers for refugees
     - income of those receiving vouchers rises from 5 to 10
     - over same time, income for comparably skilled group in host community (no vouchers) changes from 20 to 25
     - on original scale, $DID = (10 - 5) - (25 - 20) = 0\%$ effect of voucher
     - but in terms of percentage changes or logs,
       $[100] - [25] = 75$ percentage points
       $[log(10) - log(5)] - [log(25) - log(20)] = log(2) - log(1.25) = .47$
   - In general, more similarity in original outcomes is better both for this reason and to make parallel trends assumption more believable

Matching and difference-in-differences

- Parallel trends is more plausible among units that are more similar (both on X and pre-tx Y). One strategy: combine matching (or weighting) and DID
  - Match on pre-treatment covariates and (lagged) outcomes
  - Run difference-in-differences regression in matched data-set
  - Can also use inverse-propensity score weighting (Hirano, Imbens, and Ridder 2003; Imai and Kim 2012)
  - (If you match/weight so that the treated and control groups have equal means on pre-treatment Y, then DID is just Diff.)
- Randomized experiment, natural experiment, or a discontinuity can improve parallel trends assumption by asserting "$Y(0)$-trend" is balanced. So DID can be used in combination.
- This is just to say you can use the change-score rather than level of Y as your outcome in an experiment.
- With few treated units and/or many pre-treatment periods: synthetic control and related methods are applicable, but different assumptions.

“Modern” DID?

- Everything we’ve done here is to give you a foundation in “classic” DID, meaning two groups, two periods.
- Sometimes you have different units moving into treatment at different times and you still want to imagine something like parallel trends.
- People long used “two-way fixed effect” models in these cases and unfortunately still called this DID (?!)
- Then they discovered TWFE does not have a straightforward interepretation akin to DID and is troubled by either biases or confusing interpretations due to:
  - Treatment effect heterogeneity
  - Already-treated as controls
  - Event time misalignment
  - Global vs. local DID assumptions

What to do with staggered treatments etc.?

You can always run a series of separate well-constructed DID for each wave of treatment adoption, understand what each says, then average later.

See Chiu et al. paper cited above for a great review, including 6 estimators, and empirical impact, or the `fect` tutorial on 5 methods at `https://yiqingxu.org/packages/fect/05-panel.html`

An explosion of work has tried to offer different solutions. The main players you'll hear most about are...

Today's hits in place of TWFE

Approach	**What it solves and how**
Sun & Abraham (2021)	Avoids contamination by using cohort-specific event-time indicators; handles heterogeneity and staggered timing via proper reference groups.
Callaway & Sant'Anna (2021)	Computes group-time-specific ATT(g, t) with flexible, nonparametric aggregation; avoids negative weighting and uses only not-yet-treated units as controls.
Goodman-Bacon (2021)	Decomposes TWFE into 2x2 DiD comparisons; shows that bias arises from contaminated comparisons and negative weights.
de Chaisemartin & D'Haultfœuille (2020)	Aggregates clean DiD comparisons using convex weights; produces interpretable estimates even under treatment effect heterogeneity.

But remember...

What's the first-order concern with any of these?

How do newspaper endorsement affect vote choice?

Lenz and Ladd (2009) consider effect of shift in newspaper endorsements to Tony Blair in the 1997 U.K. general election

Sun, 18 March 1997

Example of Matching-plus-DID: Ladd and Lenz (2009)

Using panel data and matching techniques, we exploit a rare change in communication flows – the endorsement switch to the Labour Party by several prominent British newspapers before the 1997 United Kingdom general election—to study the persuasive power of the news media. These unusual endorsement switches provide an opportunity to test for news media persuasion while avoiding methodological pitfalls that have plagued previous studies. By comparing readers of newspapers that switched endorsements to similar individuals who did not read these newspapers, we estimate that these papers persuaded a considerable share of their readers to vote for Labour.

Difference-in-Differences Estimates

- The Sun switched endorsements. Find people who read the Sun; see if there is a change in their opinions.
- Since those who read it could have changed anyway, compare change in this group to similar people (matching) who don't read it (DID):

What is the ID assumption? Is it credible?

Difference-in-Differences Estimates

This figure shows that reading a paper that switched to Labour is associated with an $(15.2 - 6.6 =)$ 8.6 percentage point shift to Labour between the 1992 and 1997 UK elections. Paper readership is measured in the 1996 wave, before the papers switched, or, if no 1996 interview was conducted, in an earlier wave. Confidence intervals show one standard error.

Matching on 1992 Characteristics

TABLE 3 Comparing Covariates among the Treated and Untreated Groups

	All		*Difference* (Treated Minus Untreated)		
Covariates (Measured in 1992)	**Treated**	**Untreated**	**All**	**Exact**	**Genetic**
Prior Labour Vote (Labour 1, Other 0)	0.389	0.323	0.066	0.000	0.000
Prior Conservative Vote (Conservative 1, Other 0)	0.389	0.404	−0.015	0.000	0.000
Prior Liberal Vote (Liberal 1, Other 0)	0.156	0.188	−0.032	0.000	0.000
Prior Labour Party Identification (Labour 1, Other 0)	0.337	0.314	0.022	0.000	−0.005
Prior Conservative Party Identification (Conservative 1, Other 0)	0.412	0.418	−0.007	0.000	0.005
Prior Liberal Party Identification (Liberal 1, Other 0)	0.133	0.154	−0.021	0.000	0.005
Prior Labour Party Support (Strongly Favor 1 to Strongly Oppose 0)	0.488	0.462	0.025	0.000	−0.005
Prior Conservative Party Support (Strongly Favor 1 to Strongly Oppose 0)	0.524	0.522	0.003	0.000	0.005
Prior Political Knowledge (High 1, Mid .5, Low 0)	0.545	0.671	−0.126	0.000	−0.007

Difference-in-Differences in Matched Data

			Preprocessed with Matching			
			Exact on Selected Variables		**Genetic on All Variables**	
	Bivariate	**Multivariate (Probit)**	**Bivariate**	**Multivariate (Probit)**	**Bivariate**	**Multivariate (Probit)**
Among All Readers						
Treatment Effect (%)	8.6	12.2	10.9	14.0	10.4	9.6
(Standard error)	(3.0)	(3.6)	(4.1)	(6.0)	(4.3)	(4.9)
n Treated / *n* Control	211/1382	211/1382	192/192	192/192	211/211	211/211
Among Habitual Readers						
Treatment Effect (%)	12.7	23.1	17.9	23.4	15.8	25.7
(Standard error)	(4.1)	(6.4)	(5.4)	(11.3)	(6.6)	(9.0)
n Treated / *n* Control	102/1382	102/1382	95/95	95/95	102/102	102/102